

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 2

6031/2

JUNE 2019 SESSION

1 hour 30 minutes

Candidates answer on the question paper
 Additional materials:
 Data Booklet
 Scientific Electronic Calculator

TIME: 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
6	
TOTAL	

This question paper consists of 11 printed pages and 1 blank page.

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Answer all questions

1 (a) Define the term

(i) *metallic bond*,

(ii) *atomic number*.

[2]

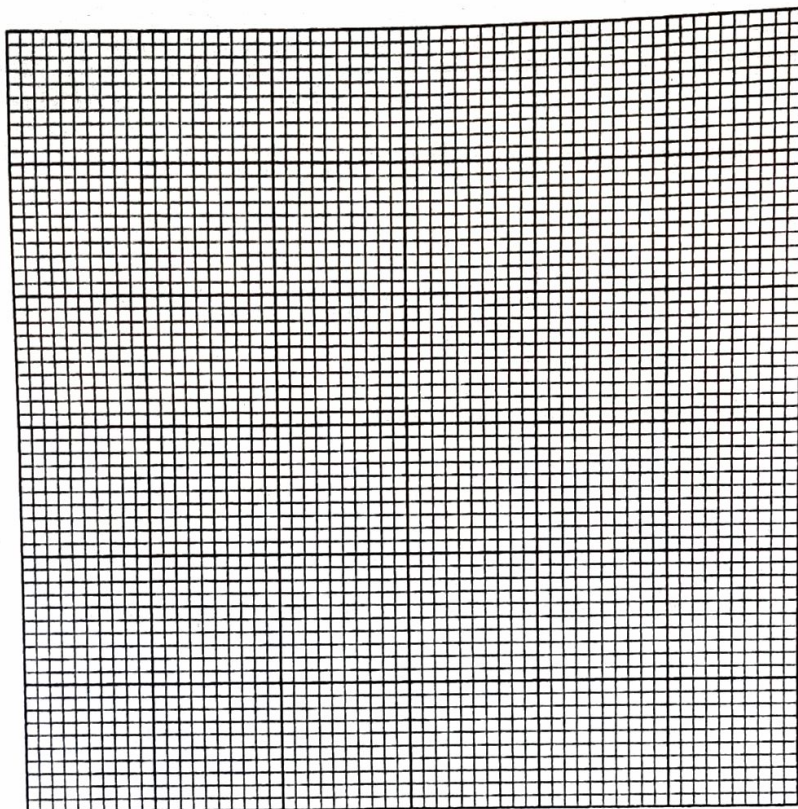
(b) Table 1.1 shows percentage (%) relative abundances and m/e values for strontium.

Table 1.1

% relative abundance	m/e
10	86
7	87
83	88

3

- (i) Draw the mass spectrum of strontium.



- (ii) Calculate the relative atomic mass of strontium.

[4]

- (c) Show how the molecule $\text{H}_2\text{NCH}_2\text{COOH}$ forms hydrogen bonds.

[2]

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Examiner's
Use

(d) Explain why graphite is used as a lubricant.

[2]

[Total 10]

2

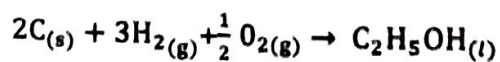
(a) (i) Define the term *standard enthalpy change of atomisation*.

(ii) Write an equation for the atomisation of ethanol.

(iii) Determine the standard enthalpy change of atomisation of ethanol.
[Use of relevant data from the data booklet is recommended]

[4]

(b) (i) Carbon reacts with hydrogen and oxygen according to the equation



Name the enthalpy change represented by the chemical change.

- (ii) Table 1.2 shows the enthalpy changes of combustion for the product and reactants in the reaction in b(i).

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Table 1.2

substance	ΔH_c^θ (kJ mol ⁻¹)
CH ₃ CH ₂ OH	- 1 367
H ₂	- 286
C (graphite)	- 1 393

Calculate ΔH_r^θ for the chemical change in b(i).

- (iii) Draw an energy reaction pathway diagram for the reaction in b (i).

[6]
[Total:10]

- 3 (a) State and explain the physical states, at s.t.p, of

- (i) CO₂,

- (ii) PbO.

- (b) (i) Write a balanced chemical equation to show how germanium reacts with chlorine.

- (ii) Describe the structure and type of bonding in the germanium compound produced in b (i).

- (iii) Comment on the boiling point of the germanium compound produced in b(i).

[4]

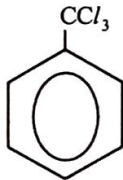
- (c) Describe and explain the variation in electrical conductivities of Group (II) elements.

[2]

[Total 10]

- 4 (a) Table 4.1 shows incomplete information about some reactions of methylbenzene with excess chlorine.

Table 4.1

condition(s) for reaction	type of reaction	structural formula of organic product
	electrophillic substitution	
		

- (i) Complete Table 4.1 to determine how methylbenzene reacts with excess chlorine under different conditions.
- (ii) Give the mechanism for the electrophilic substitution reaction.

(b) Explain why

(i) a solution of hydrogen chloride in methylbenzene is **not** acidic.

(ii) methylbenzene is insoluble in water.

[3]

[Total:10]

5 (a) Fig. 5.1 shows the structure of a flavonol, a natural organic molecule.

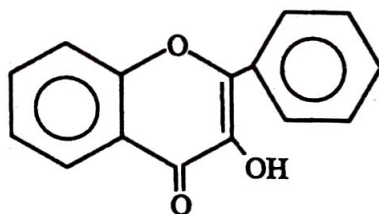


Fig. 5.1

(i) State any **three** functional groups in the flavonol.

1.

2.

3.

(ii) Describe observations made when the flavonol reacts with

1. 2,4-dinitrophenylhydrazine,

2. aqueous bromine.

[5]

(b) Draw the structure of the organic products formed when the flavonol reacts with

(i) ethanoic acid under reflux,

(ii) LiAlH_4 .

[2]

(c) Fig. 5.2 shows the structure of aspartic acid.

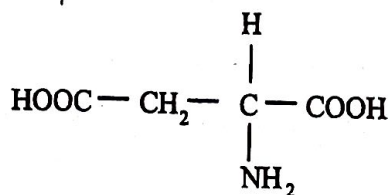


Fig. 5.2

Draw the structure of the predominant ion formed when aspartic acid is in

1. a neutral solution,

2. an alkaline solution,

3. an aqueous solution of HCl .

[3]
[Total 10]

6 (a) (i) Define the term *heterogeneous catalysis*.

(ii) Explain why transition metals are used in hydrogenation reactions.

[3]

(b) Draw diagrams to show the structures of *cis* and *trans* tetraaminedichlorochromium (III) ions.

[2]

- (c) (i) Define the term *nanotechnology*.

- (ii) State any **three** biomedical applications of nanotechnology.

1. _____

2. _____

3. _____ [5]

[Total:10]

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